

ANUJ KESHA NAYAK

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I am investigating reliable information processing in quantum systems under physical constraints from information theoretic perspective. My past academic and professional background includes information theory, digital communications, signal processing, and network science.

Personal webpage: https://nayakanuj.github.io/personal_webpage/.

List of papers: [link to google scholar page](#).

EDUCATION

University of Illinois, Urbana-Champaign, IL

June 2021 - Present

PhD in Electrical and Computer Engineering (CGPA: 4.0), Advisor: Dr. Lav Varshney.

North Carolina State University, Raleigh, NC

August 2016 - December 2018

M.S. in Electrical Engineering (CGPA: 4.0), Advisor: Dr. Huaiyu Dai.

PES Institute of Technology, Bengaluru, India

August 2010 - September 2014

B. E. in Electronics and Communication Engineering (CGPA: 8.81/10).

RESEARCH/ACADEMIC EXPERIENCE

Graduate Research Assistant, University of Illinois, Urbana-Champaign

June 2021 - Present

Quantum information processing under physical constraints

- *Iterative joint decoding and sensing under radiation-induced correlated noise* (2025-26)

Formulated joint decoding and noise (quasi-particle density) estimation under radiation-induced (muon) correlated noise in superconducting qubits as a variational inference problem, and derived iterative algorithms.

Demonstrated up to $6\times$ improvement in logical error probability of the proposed iterative algorithms (joint noise estimation and decoding) over baseline (without noise estimation) for surface codes and bivariate bicycle quantum LDPC codes through **Stim** simulations.

- *Resilience of quantum message passing* under both channel noise and circuit noise, and their effects on computational and sensing advantage (ongoing).

Fundamental limits in quantum information processing

- *Fundamental limits of reliability and storage capacity in quantum memories* (2024).

Derived asymptotic and non-asymptotic bounds on storage capacity: proved achievability using quantum expander codes, and obtained upper bounds using converse arguments from quantum channel coding theorem.

- *Limits of fault-tolerant quantum computing (FTQC) for classical problems* (2023).

Derived minimum required redundancy for FTQC using one-shot communication bounds.

Proposed provably convergent optimization algorithms to obtain fault-tolerance noise thresholds.

AI and Machine learning

- Provided a mathematical explanation for scaling laws, emergent abilities and performance plateauing in language models using tools from information theory and random network theory (2024) ([github link](#)).

Graduate Research Assistant, NC State University

May 2017 - August 2018

- Thesis: Information Spreading in Dynamic Networks ([link](#)).

- Theoretical study of information spreading in Markovian time-varying graphs.

- Opinion maximization in social networks: proposed 4 algorithms for opinion maximization through smart information spreading. Demonstrated through simulations the better performance of our centralized and distributed reinforcement learning algorithms on both synthetic and real-world graphs ([github link](#)).

PROFESSIONAL EXPERIENCE

DSP Architecture Intern, Marvell Semiconductors, Santa Clara

May 2025 - August 2025

- Designed advanced non-linear equalization algorithms for PAM4 signals in high-speed data center links.

- Implemented and optimized adaptive FFE and DFE-based architectures for ISI mitigation.

- Achieved >0.5 dB SNR gain and $>30\%$ BER improvement over the baseline on ADC samples from measured signal traces, with simulations developed in **PyTorch**.

- Patent application submitted for the developed equalization methods.

Modem Systems Engineer, Wireless R&D, Qualcomm, San Diego, CA January 2019 - May 2021

- 5G-NR Uplink Tx evaluator: phase/frequency offset compensation, PRACH detector and MIMO equalizer.
- SmartTx evaluator in Matlab for maximum permissible radiation exposure control.
- Signal processing lab support to software/test teams for Snapdragon X55 modem bring-up.

Design Engineer (Algorithm), Signalchip Innovations, Bengaluru, India August 2014 - June 2016

- LTE and WCDMA Downlink Receiver - Designed fixed point algorithms for cell search, channel equalization, demodulation, phase/frequency synchronization.
- Worked with hardware (FPGA/ASIC) and firmware teams for design revisions, implementation and validation.
- Large-scale simulations (small-scale fading channels with doppler and additive noise) using HPCs to ensure compliance of algorithms with 3GPP specifications.

RELEVANT COURSEWORK

Manipulating Quantum Systems (ECE 498), Quantum Information Processing Theory, Quantum Channels, Random processes, Introduction to Optimization, Generative AI Models, Statistical Reinforcement Learning, Information Theory, Detection and Estimation Theory, Probabilistic Graphical Modeling, Signal Processing for Communications and Networking.

SELECTED PUBLICATIONS

Link to Google scholar page for the full list of papers: [google scholar link](#).

- A. K. Nayak, P. G. Baity, P. J. Love, N. Jeon, B. J. Yoon, A. Hoisie, and L. R. Varshney, “**Iterative decoding of stabilizer codes under radiation-induced correlated noise**,” arXiv:2603.18231 (2026).
- A. K. Nayak, E. Chitambar, and L. R. Varshney, “**Reliable Quantum Memories with Unreliable Components**,” Physical Review A 110.3 (2024): 032423.
- Uthirakalyani. G, A. K. Nayak, A. Chatterjee, and L. R. Varshney, “**Limits of fault tolerance on resource-constrained quantum circuits for classical problems**,” Phys. Rev. A, vol. 108, p. 052425, Nov 2023.
- Uthirakalyani. G, A. K. Nayak, and A. Chatterjee, “**A Converse for Fault-tolerant Quantum Computation**,” Quantum 7 (2023): 1087.
- P. G. Baity, A. K. Nayak, L. R. Varshney, N. Jeon, B. Yoon, P. J. Love, and A. Hoisie, “**Characterizing Quantum Error Correction Performance of Radiation-induced Errors**,” arXiv:2602.06202 (2026).
- A. K. Nayak, L. R. Varshney, “An Information Theory of Compute-Optimal Size Scaling, Emergence, and Plateaus in Language Models,” IEEE Journal of Selected Topics in Signal Processing 19.7 (2025): 1338-1348.
- A. R. Ellis-Mohr, A. K. Nayak, and L. R. Varshney. ”A theory of inference compute scaling: reasoning through directed stochastic skill search.” Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences 384.2315 (2026).
- A. Wadell, A. Bhutani, V. Azumah, A. R. Ellis-Mohr, C. Kelly, H. Zhao, A. K. Nayak, K. Hegazy, A. Brace, H. Lin, M. Emani, V. Vishwanath, K. Gering, M. Alkan, T. Gibbs, J. Wells, L. R. Varshney, B. Ramsundar, K. Duraisamy, M. W. Mahoney, A. Ramanathan, V. Viswanathan. ”Foundation models for discovery and exploration in chemical space.” arXiv:2510.18900 (2025).
- A. K. Nayak, S. Hosseinalipour, and H. Dai, “Smart Information Spreading for Opinion Maximization in Social Networks.” IEEE International Conference on Computer Communications (INFOCOM), 2019.

INVITED TALKS AND PRESENTATIONS

- A. R. Ellis-Mohr, A.K. Nayak, “A Theory for Training and Inference Compute Scaling Paradigms,” IBM Research, Virtual, May 2025.
- “An Information Theory of Training Compute Scaling in Language Models,” Sanjeev Arora’s group, Princeton University, Virtual, October 2024.
- “An Information Theory of Compute Optimal Size Scaling, Emergence and Plateaus in Language Models”, selected for Oral Presentation at Neural Compression Workshop, NeurIPS 2024.
- “Reliable Quantum Memories,” (Poster), QIS summer school, Oak Ridge National Lab, July 2024.
- “Fundamental Limits of FTQC for Classical Problems (Poster),” QIS summer school, Fermilab, July 2023.

CODING SKILLS

Python, Qiskit, PyTorch, Matlab, Version control (github and SVN), C, C++.